

Finite-Control-Set Model Predictive Control for DFIG Wind Turbines

Peng Kou, *Member, IEEE*, Deliang Liang, *Senior Member, IEEE*, Jing Li, Lin Gao, and Qiji Ze

Abstract—This paper presents a time efficient finite-control-set model predictive control (FCS-MPC) scheme for the doubly fed induction generator system. In this scheme, the switching states of the rotor side converter are directly taken as control inputs. This way, the optimized control action can be directly applied to the converter. Compared with the existing FCS-MPC approaches, the salient feature of the proposed scheme is the reduction of the computation time. By introducing a set of augmented decision variables, the original intractable binary quadratic programming problem in FCS-MPC can be analytically transformed to a binary linear programming problem, which can be solved efficiently. By this means, the computation time of the proposed scheme is much less than that of the existing schemes. This reduction in computation time enables FCS-MPC with longer prediction horizons, thus yielding better control performance.

Note to Practitioners—This paper was motivated by the problem of improving the control performance for the doubly fed induction generator (DFIG). Because of its advantages, such as high energy efficiency, low acoustic noise, variable speed operation, and reduced converter rating, DFIG has emerged as a promising solution for the wind power generation. However, existing DFIG controllers have limitations, such as difficult to tune the parameters, inefficient to handle constraints, and do not consider the discrete operation of the power converters. To overcome these limitations, this paper proposes a new DFIG control scheme based on finite-control-set model predictive control (FCS-MPC), which has the abilities of online optimal control, explicitly handling constraints, and directly generating the switching signals for the power converters. A computationally efficient algorithm is proposed to reformulate the FCS-MPC problem as a linear program, thereby significantly reducing the computational efforts and rendering the proposed scheme more practical for implementation. Simulation results validate the effectiveness of the proposed control scheme. This scheme can be implemented in the field-programmable gate array, which will be our future work.

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P. Kou, D. Liang, L. Gao, and Q. Ze are with the State Key Laboratory of Electrical Insulation and Power Equipment, Shaanxi Key Laboratory of Smart Grid, School of Electrical Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: kou-peng@mail.xjtu.edu.cn).

J. Li is with the Academy of Space Information System, Xi'an 710100, China.

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Index Terms—Doubly fed induction generator (DFIG), finite control set (FCS), linear programming, model predictive control (MPC), wind energy.

I. INTRODUCTION

THE last decade has witnessed an impressive development of renewable energy technologies. Of the many renewable generation resources, wind generation is considered as the most promising one and hence receiving worldwide attentions [1]. Among different types of wind energy conversion systems, the doubly fed induction generator (DFIG)-based wind turbines have arisen as a commonly used solution, due to its advantages, such as high energy efficiency, diminish acoustic noise, variable speed operation, and reduced converter rating. From the view of the power system operation, it is desirable to regulate the active and reactive power of the DFIG under various wind conditions. Therefore, the control issues of the DFIG are of great importance to be properly investigated.

Currently, these control issues are commonly addressed using the vector control scheme [2], [3]. In this scheme, the DFIG rotor current is decoupled into active and reactive power components in the $d-q$ reference frame. Thus, the active and reactive power control of DFIG can be achieved by a rotor current controller, which regulates the decoupled rotor currents through the rotor side converter (RSC). Traditionally, this rotor current controller is designed using the proportional-integral-differential (PID) method. However, for DFIG control applications, conventional PID controllers have intrinsic limitations. To overcome these limitations, various advanced control schemes have been proposed for DFIG, e.g., neural network control [4], direct power control (DPC) [5], robust control [6], inner control [7], multimode control [8], and fuzzy control [9].

In general, the control objective for DFIG is twofold. The primary objective is to track the desired active and reactive power references. Since the wind speed is perpetually varying, the DFIG output power has to be adjusted continuously to track the references, thus giving rise to an online control problem. The secondary objective is to limit the rotor current during the grid voltage dip. Due to this objective, the DFIG control problem is a constrained one. Comprehensively considering these two objectives, the control of DFIG can be regarded as an online constrained control problem. Model predictive control (MPC) is an ideal framework to tackle this problem, since it repeatedly optimizes performance over a receding horizon and explicitly handles the system constraints [10]. Because of these benefits, there

have already been some applications of MPC to the control of DFIG [11]–[15].

MPC can be widely categorized into continuous control set and finite control set (FCS) methods. Among these two types of methods, the FCS-MPC is particularly suitable for the control of power converters [16]–[19]. The key feature of FCS-MPC is that it directly uses the gating signals of the power switches (0: OFF, 1: ON) as control inputs. Therefore, FCS-MPC controller can directly output a switching state to the converter, without requiring a modulation stage. In this sense, compared with the continuous MPC, FCS-MPC is more suitable for the control of DFIG.

However, extending the FCS-MPC to the control of DFIG is not straightforward. A major obstacle is the large computation time requirement. In the existing FCS-MPC schemes, the cost function generally results in a binary quadratic programming (BQP) problem, which is analytically intractable. Therefore, the optimization of the control action is commonly achieved by exhaustive search, which is very time consuming [17]. Such long computation time would cause two problems. On one hand, it will exhaust the controller's computation limit, making operation at high sampling frequency impossible. On the other hand, it limits the length of the prediction horizon to only one or two time steps [18], [19], due to the fact that the computation time increases exponentially with respect to the length of the prediction horizon. This in turn limits the control performance, since in the MPC framework, longer prediction horizons generally yield better control performance [16]. Due to these limitations, to the best of our knowledge, although there have been some applications of FCS-MPC to power converters and electric drive systems [20]–[28], few attempt to apply FCS-MPC to DFIG, all the aforementioned MPC schemes for DFIG [11]–[15] are based on the continuous MPC.

In order to overcome the above-mentioned limitations, this paper introduces a new time efficient FCS-MPC scheme for the DFIG system. The key technical contribution of this paper can be summarized as follows.

- 1) The proposed FCS-MPC scheme has the abilities of online optimal control, explicitly handling constraints, and directly generating the switching signals for the RSC of DFIG.
- 2) By using the augmented decision variables, the intractable BQP problem in FCS-MPC is equivalently transformed to a binary linear programming (BLP) problem, which can be solved efficiently, so the computation time is significantly reduced.
- 3) The reduction in computation time enables FCS-MPC with longer prediction horizons, thus yielding better control performance.

The remainder of this paper is organized as follows. Section II presents the mathematical model of the considered DFIG system. Section III gives the precise control problem formulation and the detailed description of the proposed FCS-MPC scheme. Section IV contains the description of the simulations and the discussion of the results. Finally, Section V draws conclusions.

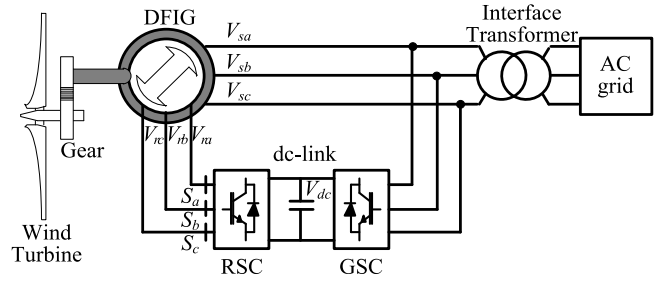


Fig. 1. Schematic of a grid-connected DFIG system.

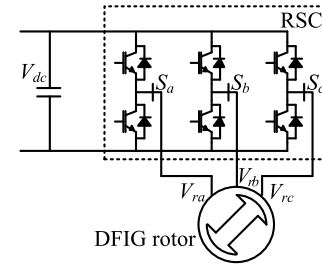


Fig. 2. Topology of RSC.

II. SYSTEM DESCRIPTION AND MODELING

A. System Layout

The schematic of a grid-connected DFIG system is shown in Fig. 1. This system is composed of a wind turbine, a gear-box, a DFIG generator, an RSC, a direct current (dc) link, as well as a grid side converter (GSC). In Fig. 1, V_{dc} denotes the dc-link voltage, S_a , S_b , and S_c denote the ON and OFF states of the power switches in the RSC, V_{sa} , V_{sb} , and V_{sc} denote the stator voltages of the DFIG, while V_{ra} , V_{rb} , and V_{rc} denote the three-phase voltages applied to the DFIG rotor, respectively.

In the DFIG system, the main task of the RSC is to control the DFIG's stator active and reactive power outputs, whereas the GSC is responsible for regulating the dc-link voltage. In this paper, we focus on the DFIG power control, so we model the RSC and the DFIG generator in the Sections II-B and II-C.

B. RSC Model

Fig. 2 shows the topology of the RSC. Here, we use a vector $\mathbf{S} = [S_a, S_b, S_c]^T$ to denote the switching state of the RSC, the definitions of V_{dc} , V_{ra} , V_{rb} , and V_{rc} are the same as those in Fig. 1.

In the RSC, the switching state of each leg can be expressed as follows:

$$S_x = \begin{cases} 1, & \text{if the } x\text{-th leg upper switch is ON} \\ 0, & \text{if the } x\text{-th leg lower switch is ON.} \end{cases} \quad (1)$$

Here, x is the index of the leg.

Using the switching state functions, the **phase voltages** of the rotor can be written as

$$\begin{bmatrix} V_{ra} \\ V_{rb} \\ V_{rc} \end{bmatrix} = \frac{V_{dc}}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \mathbf{S}. \quad (2)$$

As will be seen in the next section, the proposed FCS-MPC scheme is based on the **vector control principle**. Therefore, the

phase voltage of the rotor has to be transformed to the $d-q$ reference frame, as follows:

$$\begin{bmatrix} V_{rd} \\ V_{rq} \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos(\theta) & \cos\left(\theta - \frac{2}{3}\pi\right) & \cos\left(\theta + \frac{2}{3}\pi\right) \\ -\sin(\theta) & -\sin\left(\theta - \frac{2}{3}\pi\right) & -\sin\left(\theta + \frac{2}{3}\pi\right) \end{bmatrix} \times \begin{bmatrix} V_{ra} \\ V_{rb} \\ V_{rc} \end{bmatrix} \quad (3)$$

where V_{rd} and V_{rq} are the d -axis and q -axis rotor voltages, θ is the rotor flux angle.

Combining (1)–(3), we can obtain the relationship between the RSC switching state and the DFIG rotor voltage vector, which is given by

$$\begin{bmatrix} V_{rd} \\ V_{rq} \end{bmatrix} = V_{dc} \mathbf{C}_{sr} \mathbf{C}_{ss} \mathbf{S} \quad (4)$$

with

$$\mathbf{C}_{ss} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

$$\mathbf{C}_{sr} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos(\theta) & \cos\left(\theta - \frac{2}{3}\pi\right) & \cos\left(\theta + \frac{2}{3}\pi\right) \\ -\sin(\theta) & -\sin\left(\theta - \frac{2}{3}\pi\right) & -\sin\left(\theta + \frac{2}{3}\pi\right) \end{bmatrix}.$$

In the proposed FCS-MPC scheme, the switching state $\mathbf{S}$ will be used as the control input, as explained in Section II-D.

C. DFIG Generator Model

By combining the rotor voltage and flux equations, we can get the dynamic model of the DFIG [11], as follows:

$$\begin{aligned} \frac{di_{rd}}{dt} &= -\frac{R_r}{L_r} i_{rd} + w_r i_{rq} + \frac{1}{L_r} V_{rd} \\ \frac{di_{rq}}{dt} &= -w_r i_{rd} - \frac{R_r}{L_r} i_{rq} + \frac{1}{L_r} V_{rq} - \frac{L_m V_s}{L_r L_s w_s} w_r. \end{aligned} \quad (5)$$

Here, i_{rd} and i_{rq} are the d -axis and q -axis rotor currents, R_r is the rotor resistance, L_s and L_r are the stator and rotor inductances, L_m is the mutual inductance, V_s is the stator voltage, and w_r and w_s are the rotor and stator electrical frequencies. $w_s = 2\pi f$ is a constant value, since $f = 60$ Hz is fixed by the grid. The rotor electrical frequency w_r must satisfy the slip frequency constraint, that is

$$N_p w_m = w_s - w_r \quad (6)$$

where N_p is the number of pole-pairs and w_m is the rotor mechanical angular speed.

In our setting, the output of interest is the stator active and reactive power. In the $d-q$ reference frame, they can be calculated as

$$P = -\frac{L_m V_s}{L_s} i_{rq}, \quad Q = \frac{V_s^2}{L_s w_s} - \frac{L_m V_s}{L_s} i_{rd}. \quad (7)$$

Equations (5)–(7) constitute the model of DFIG. This model includes the nonlinear terms $w_r i_{rd}$ and $w_r i_{rq}$, and thus, needs to be linearized.

In order to linearize the model (5)–(7), in this paper, the rotor mechanical speed w_m is assumed to be constant for one control cycle. This is a valid approximation, due to the fact that in one control cycle, the mechanical time constant is, in general, much greater than the electrical time constants [11], [29]. Consequently, the rotor electrical frequency w_r can also be assumed constant in each control cycle, due to the fact that w_s is fixed by the grid and $w_r = w_s - N_p w_m$. In this way, the dynamics of i_{rd} and i_{rq} are decoupled from w_r . As a result, the nonlinear model (5)–(7) becomes a linear model.

Once the dynamic model is linearized, we can further obtain a discrete-time state-space model of DFIG via the exponential discretization technique, as follows:

$$\begin{aligned} \begin{bmatrix} i_{rd}(k+1) \\ i_{rq}(k+1) \end{bmatrix} &= \mathbf{A} \begin{bmatrix} i_{rd}(k) \\ i_{rq}(k) \end{bmatrix} + \mathbf{B}_V \begin{bmatrix} V_{rd}(k) \\ V_{rq}(k) \end{bmatrix} \\ &\quad - \begin{bmatrix} 0 \\ T \end{bmatrix} \left(\frac{L_m V_s w_r}{L_r L_s w_s} \right) \\ \begin{bmatrix} Q(k) \\ P(k) \end{bmatrix} &= \begin{bmatrix} -L_m V_s / L_s & 0 \\ 0 & -L_m V_s / L_s \end{bmatrix} \begin{bmatrix} i_{rd}(k) \\ i_{rq}(k) \end{bmatrix} \\ &\quad + \begin{bmatrix} V_s^2 / (L_s w_s) \\ 0 \end{bmatrix} \end{aligned} \quad (8)$$

with

$$\begin{aligned} \mathbf{A} &= \exp \left(\begin{bmatrix} -R_r T / L_r & w_r T \\ -w_r T & -R_r T / L_r \end{bmatrix} \right) \\ \mathbf{B}_V &= \begin{bmatrix} \frac{R_r}{L_r} & w_r \\ -w_r & -\frac{R_r}{L_r} \end{bmatrix}^{-1} \left(\exp \left(\begin{bmatrix} -\frac{R_r T}{L_r} & w_r T \\ -w_r T & -\frac{R_r T}{L_r} \end{bmatrix} \right) - \mathbf{I} \right) \\ &\quad \times \begin{bmatrix} \frac{1}{L_r} & 0 \\ 0 & \frac{1}{L_r} \end{bmatrix}. \end{aligned}$$

Here, T is the sampling period and k denotes the time step.

D. Combined Model of DFIG and RSC

Since FCS-MPC directly manipulates the RSC switch positions, $\mathbf{S} = [S_a, S_b, S_c]^T$ should be taken as the control inputs. This can be achieved by combining the RSC model (4) and the DFIG model (8). By this means, the combined model of DFIG and RSC can be written as

$$\begin{aligned} \underbrace{\begin{bmatrix} i_{rd}(k+1) \\ i_{rq}(k+1) \end{bmatrix}}_{\mathbf{x}(k+1)} &= \underbrace{\mathbf{A} \begin{bmatrix} i_{rd}(k) \\ i_{rq}(k) \end{bmatrix}}_{\mathbf{x}(k)} + \underbrace{V_{dc} \mathbf{B}_V \mathbf{C}_{sr} \mathbf{C}_{ss}}_{\mathbf{B}} \underbrace{\mathbf{S}}_{\mathbf{u}(k)} \\ &\quad + \underbrace{\begin{bmatrix} 0 \\ T \end{bmatrix}}_{\mathbf{e}} \underbrace{\left(-\frac{L_m V_s w_r}{L_r L_s w_s} \right)}_{z(k)} \\ \underbrace{\begin{bmatrix} Q(k) \\ P(k) \end{bmatrix}}_{\mathbf{y}(k)} &= \underbrace{\begin{bmatrix} -L_m V_s / L_s & 0 \\ 0 & -L_m V_s / L_s \end{bmatrix}}_{\mathbf{C}} \underbrace{\begin{bmatrix} i_{rd}(k) \\ i_{rq}(k) \end{bmatrix}}_{\mathbf{x}(k)} \\ &\quad + \underbrace{\begin{bmatrix} V_s^2 / (L_s w_s) \\ 0 \end{bmatrix}}_{\mathbf{r}}. \end{aligned} \quad (9)$$

III. FCS-MPC-BLP SCHEME FOR THE DFIG SYSTEM

A. FCS-MPC Problem Formulation

In this paper, the controller is designed via the FCS-MPC approach. In this approach, a cost function has to be developed to find the optimal control actions. As discussed in Section I, the primary objective of a DFIG controller is to track the desired active and reactive power references. That is, to bring the DFIG active power output P and reactive power output Q (i.e., $\mathbf{y}$) to the set points P_{ref} and Q_{ref} (i.e., $\mathbf{y}_{\text{ref}}$), respectively. In view of this objective, at each time step k , the cost function for FCS-MPC can be designed as

$$\min_{\mathbf{U}(k)} J(k) = \|\mathbf{Y}(k) - \mathbf{Y}_{\text{ref}}(k)\|_{\mathbf{W}_\beta}^2 + \|\mathbf{U}(k)\|_{\mathbf{W}_{1-\beta}}^2 \quad (10)$$

where $\mathbf{Y}(k) = [\mathbf{y}(k+1|k), \mathbf{y}(k+2|k), \dots, \mathbf{y}(k+H|k)]^T$, $\mathbf{Y}_{\text{ref}}(k) = [\mathbf{y}_{\text{ref}}(k+1|k), \mathbf{y}_{\text{ref}}(k+2|k), \dots, \mathbf{y}_{\text{ref}}(k+H|k)]^T$, here, $(k+h|k)$ denotes the value predicted for look ahead step $k+h$ based on the information available at time step k . $\mathbf{U}(k)$ denotes the control sequence at time step k , i.e., $\mathbf{U}(k) = [\mathbf{u}(k), \mathbf{u}(k+1), \dots, \mathbf{u}(k+H-1)]^T$. $\mathbf{W}_\beta = \text{diag}(\beta)$ and $\mathbf{W}_{1-\beta} = \text{diag}(1-\beta)$ are weighting matrices between two optimization objectives. **For simplicity, both prediction and control horizons are set to H .**

In the cost function (10), the output prediction $\mathbf{y}(k+h|k)$ can be computed recursively from the state-space model (9). That is, we can get an explicit expression between $\mathbf{y}(k+h|k)$ and control sequence $[\mathbf{u}(k), \mathbf{u}(k+1), \dots, \mathbf{u}(k+h-1)]^T$, which is given by

$$\begin{aligned} \mathbf{y}(k+h|k) = & \mathbf{CA}^h \mathbf{x}(k) + \mathbf{CA}^{h-1} (\mathbf{Bu}(k) + \mathbf{Ez}(k)) \\ & + \mathbf{CA}^{h-2} (\mathbf{Bu}(k+1|k) + \mathbf{Ez}(k+1|k)) + \dots \\ & + \mathbf{CA}^1 (\mathbf{Bu}(k+h-1|k) + \mathbf{Ez}(k+h-1|k)) + \mathbf{r}. \end{aligned} \quad (11)$$

This prediction process is a recursive function, which calls itself until the look ahead step h is equal to the prediction horizon H .

Besides the power tracking objective, the grid usually requires the DFIG to have fault ride through capability during the grid fault, e.g., grid voltage dip or swell fault. In these fault conditions, **the over current on the RSC is easy to occur**, due to the excessive power coming from the rotor side [12]. In light of this, a secondary control objective is to limit the rotor over current during grid fault conditions. **To achieve this, we impose constraints on the rotor currents i_{rd} and i_{rq}** , as follows:

$$-i_{rd,\max} \leq i_{rd}(k+h|k) \leq i_{rd,\max}, \quad h = 1, 2, \dots, H \quad (12)$$

$$-i_{rq,\max} \leq i_{rq}(k+h|k) \leq i_{rq,\max}, \quad h = 1, 2, \dots, H. \quad (13)$$

The cost function (10) and the constraints (12) and (13) form the FCS-MPC optimization problem for a DFIG system.

In order to facilitate the following derivation, we **recast** this optimization problem in a matrix form. To achieve this, we

introduce the following notations:

$$\begin{aligned} \tilde{\mathbf{C}} &= \begin{bmatrix} \mathbf{C} & & \\ & \ddots & \\ & & \mathbf{C} \end{bmatrix}, \quad \Psi = \begin{bmatrix} \mathbf{B} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{AB} & \mathbf{B} & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}^{H-1}\mathbf{B} & \mathbf{A}^{H-2}\mathbf{B} & \dots & \mathbf{B} \end{bmatrix} \\ \mathbf{R} &= \begin{bmatrix} \mathbf{r} \\ \mathbf{r} \\ \vdots \\ \mathbf{r} \end{bmatrix}, \quad \Phi = \tilde{\mathbf{A}}\mathbf{x}(k) + \begin{bmatrix} \mathbf{E} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{AE} & \mathbf{E} & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}^{H-1}\mathbf{E} & \mathbf{A}^{H-2}\mathbf{E} & \dots & \mathbf{E} \end{bmatrix} \mathbf{Z} \\ \mathbf{z} &= \begin{bmatrix} z(k+1|k) \\ z(k+2|k) \\ \vdots \\ z(k+H|k) \end{bmatrix}, \quad \mathbf{I}_{rd,\max} = \begin{bmatrix} i_{rd,\max} \\ i_{rd,\max} \\ \vdots \\ i_{rd,\max} \end{bmatrix} \\ \mathbf{I}_{rq,\max} &= \begin{bmatrix} i_{rq,\max} \\ i_{rq,\max} \\ \vdots \\ i_{rq,\max} \end{bmatrix} \\ \mathbf{I}_1 &= \underbrace{[1 \quad 0 \quad 1 \quad 0 \quad \dots \quad 1 \quad 0 \quad 1 \quad 0]}_{\text{totally } 2H \text{ elements}} \\ \mathbf{I}_2 &= \underbrace{[0 \quad 1 \quad 0 \quad 1 \quad \dots \quad 0 \quad 1 \quad 0 \quad 1]}_{\text{totally } 2H \text{ elements}} \\ \mathbf{W} &= \beta(\tilde{\mathbf{C}}\Psi)^T \tilde{\mathbf{C}}\Psi + (1-\beta)\mathbf{I}_{3H \times 3H} \\ \mathbf{V} &= \beta((\tilde{\mathbf{C}}\Psi)^T (\tilde{\mathbf{C}}\Phi + \mathbf{R}) - (\tilde{\mathbf{C}}\Psi)^T \mathbf{Y}_{\text{ref}}) \\ \mathbf{L} &= \begin{bmatrix} \mathbf{I}_1 \Psi \\ -\mathbf{I}_1 \Psi \\ \mathbf{I}_2 \Psi \\ -\mathbf{I}_2 \Psi \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} \mathbf{I}_{rd,\max} - \mathbf{I}_1 \Phi \\ -\mathbf{I}_{rd,\max} + \mathbf{I}_1 \Phi \\ \mathbf{I}_{rq,\max} - \mathbf{I}_2 \Phi \\ -\mathbf{I}_{rq,\max} + \mathbf{I}_2 \Phi \end{bmatrix}. \end{aligned} \quad (14)$$

Here, $\mathbf{I}_{n \times m}$ denotes the unit $n \times m$ matrix.

Using these notations, the FCS-MPC problem (10)–(13) can be rewritten in a compact matrix form, which is given by

$$\min_{\mathbf{U}(k)} \mathbf{U}(k)^T \mathbf{W} \mathbf{U}(k) + 2\mathbf{U}(k)^T \mathbf{V} \quad (15a)$$

$$\text{s.t. } \mathbf{L} \mathbf{U}(k) \leq \mathbf{K}. \quad (15b)$$

From (15), we see that the FCS-MPC problem for a DFIG system can be formulated as a constrained quadratic programming problem. The decision variables of this problem are switching states $\mathbf{U}(k)$. Since the value of $\mathbf{U}(k) = [S_a(k), S_b(k), S_c(k)S_a(k+1), S_b(k+1), S_c(k+1), \dots, S_a(k+H-1), S_b(k+H-1), S_c(k+H-1)]^T$ are restricted to be binary, (15) is actually a BQP problem, which cannot be solved efficiently. In the previous FCS-MPC works, such problems are generally solved by exhaustive search. That is, in each control cycle, the controller evaluates all possible 2^{3H} combinations of $S_a(k), S_b(k), S_c(k)S_a(k+1), S_b(k+1), S_c(k+1), \dots, S_a(k+H-1), S_b(k+H-1), S_c(k+H-1)$, and then selects the one that minimizes the cost function (15a). Such an exhaustive search is very time consuming, especially when the prediction horizon H is large.

B. Solving the FCS-MPC Optimization Problem Using BLP

As presented in the above section, the formulation of FCS-MPC results in a BQP problem, which cannot be solved efficiently. **In order to address this issue, we propose to convert this problem to an equivalent BLP problem**, which can be readily solved using implicit enumeration techniques.

To do so, we first reindex the decision variables $\mathbf{U}(k) = [S_a(k), S_b(k), S_c(k), S_a(k+1), S_b(k+1), S_c(k+1), \dots, S_a(k+H-1), S_b(k+H-1), S_c(k+H-1)]^T$ as

$$\mathbf{U}(k) = [U_1, U_2, U_3, \dots, U_{3H-1}, U_{3H}]. \quad (16)$$

In addition, we introduce an augmented vector of decision variables, which is given by

$$\begin{aligned} \tilde{\mathbf{u}} = & [U_1^2, U_2^2, U_3^2, \dots, U_{3H-1}^2, U_{3H}^2, \\ & U_1 U_2, U_1 U_3, \dots, U_1 U_{3H-1}, U_1 U_{3H}, \\ & U_2 U_3, U_2 U_4, \dots, U_2 U_{3H-1}, U_2 U_{3H}, \\ & \vdots \\ & U_{3H-1} U_{3H}] \\ = & [\{U_i^2\} \cup \{U_i U_j\}]. \quad j = 1, 2, \dots, 3H, i \neq j. \end{aligned} \quad (17)$$

The decision variables in (17) can also be reindexed as

$$\tilde{\mathbf{u}} = [\tilde{u}_1, \tilde{u}_2, \tilde{u}_3, \dots, \tilde{u}_{N-1}, \tilde{u}_N]. \quad (18)$$

Here, $N = 3H + C_2^{3H}$ denotes the number of augmented decision variables.

Using the notation $\tilde{\mathbf{u}}$, the original BQP cost function (15a) can be represented as a linear sum of the augmented decision variables, together with a new set of constraints, as follows:

$$\min_{\tilde{\mathbf{u}}} \tilde{\mathbf{w}} \tilde{\mathbf{u}}^T \quad (19a)$$

$$\text{s.t. } \tilde{u}_i + \tilde{u}_j \leq U_i U_j, \quad i, j = 1, 2, \dots, 3H, i < j \quad (19b)$$

$$\tilde{u}_i \geq U_i U_j, \quad i, j = 1, 2, \dots, 3H, i \neq j \quad (19c)$$

where

$$\begin{aligned} \tilde{\mathbf{w}} = & [\mathbf{W}_{1,1}, \mathbf{W}_{2,2}, \dots, \mathbf{W}_{3H,3H}, \mathbf{W}_{1,2}, \mathbf{W}_{1,3}, \dots, \mathbf{W}_{1,3H}, \\ & \mathbf{W}_{2,3}, \mathbf{W}_{2,4}, \dots, \mathbf{W}_{2,3H}, \\ & \vdots \\ & \mathbf{W}_{3H-1,3H}] \\ = & [\{\mathbf{W}_{i,j}\}], \quad i, j = 1, 2, \dots, 3H, i \leq j. \end{aligned} \quad (20)$$

The constraint set (19b) consists of C_2^{3H} constraints, while (19c) consists of $3H(3H-1)$ constraints.

In (16), the value of U_i is restricted to be binary, so we have

$$U_i = \tilde{u}_i, \quad i = 1, 2, \dots, 3H. \quad (21)$$

Considering this fact, the original rotor current constraints (12) and (13) can be rewritten as

$$\tilde{\mathbf{L}} \tilde{\mathbf{u}} \leq \mathbf{K} \quad (22)$$

where

$$\tilde{\mathbf{L}} = [\mathbf{L} | \mathbf{0}_{4H \times C_2^{3H}}]. \quad (23)$$

Here, $\mathbf{0}_{n \times m}$ denotes the all-zero $n \times m$ matrix.

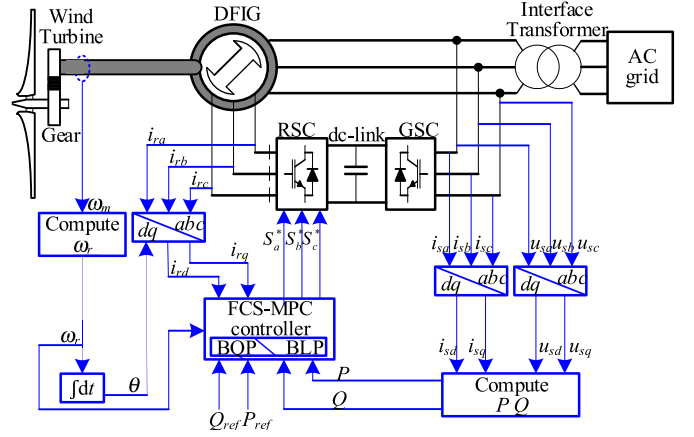


Fig. 3. Schematic of the proposed FCS-MPC-BLP for a DFIG system.

Collecting (19) and (22), we obtain a new binary programming problem, which is given by

$$\min_{\tilde{\mathbf{u}}} \tilde{\mathbf{w}} \tilde{\mathbf{u}}^T \quad (24a)$$

$$\text{s.t. } \tilde{u}_i + \tilde{u}_j \leq U_i U_j, \quad i, j = 1, 2, \dots, 3H, i < j \quad (24b)$$

$$\tilde{u}_i \geq U_i U_j, \quad i, j = 1, 2, \dots, 3H, i \neq j \quad (24c)$$

$$\tilde{\mathbf{L}} \tilde{\mathbf{u}} \leq \mathbf{K}. \quad (24d)$$

It is clear that the cost function (24a) and the constraints (24b) and (24d) are all linear in $\tilde{\mathbf{u}}$. Therefore, this problem is a standard BLP with linear constraints, which can be easily solved using standard and highly efficient Branch-and-Bound method software tools [30].

In the BLP problem (24), there are totally $3H + C_2^{3H}$ decision variables and $3H(3H-1) + C_2^{3H}$ linear constraints. Although the number of decision variables and constraints in the BLP problem (24) are more than those in the BQP problem (15), solving the BLP problem (24) requires much shorter computation time. This is confirmed by the simulation results in Section IV-C. Furthermore, since the computation time is reduced, we can use longer prediction horizons, thus obtaining better control performance. This is validated by the simulation results in Section IV-B.

From the above-mentioned analysis, we see that the BLP problem (24) is equivalent with the original FCS-MPC optimization problem. Therefore, at every time step k , solving problem (24) yields an optimal control sequence $\mathbf{U}^*(k)$. The first control action $\mathbf{u}(k)^* = [S_a^*(k), S_b^*(k), S_c^*(k)]^T$ from this sequence is applied to the RSC. At the next time step, this procedure is repeated using the new actual system measurements, e.g., i_{rd} , i_{rq} , P , Q , and w_m . By this receding horizon approach, feedback compensation is introduced into the controller design. To facilitate the subsequent analysis, we denote the proposed scheme as FCS-MPC-BLP. The complete control diagram of the proposed FCS-MPC-BLP scheme is shown in Fig. 3.

The proposed FCS-MPC-BLP scheme belongs to the class of FCS-MPC approaches. The strict stability for this class of predictive control technique still remains as an open problem [31]. However, as stated in [31], the practical stability of FCS-MPC can be proved. Here, the term “practical” is used to emphasize that only stability to a neighborhood of the

TABLE I
DFIG PARAMETERS

Symbol	Description	Value
P_{norm}	rated power	149.2 (kVA)
ω_s	stator electrical frequency	$2\pi \cdot 60$ (rad/s)
R_s	stator resistance	0.02475 (Ω)
R_r	rotor resistance	0.0133 (Ω)
L_m	mutual inductance	0.01425 (H)
L_r	stator inductance	0.0145(H)
L_s	rotor inductance	0.0145(H)
V_s	RMS stator voltage	575 (V)
V_{dc}	dc-link voltage	600 (V)
N_p	number of pole-pairs	2
J_r	rotor inertia	2.6 Kg·m ²

reference can be guaranteed. A proof of this remarkable conclusion can be found in this paper by Aguilera and Quevedo [31]. For the sake of completeness and the convenience of the reader, here, we give a brief discussion of this issue. Denote $a_1 = \lambda_{\min}(\text{diag}(\beta))$, $a_2 = \lambda_{\max}(\text{diag}(\beta))$, $a_3 = |\mathbf{B}^T \text{diag}(\beta) \mathbf{B} + \text{diag}(1 - \beta)|$, and $\rho = 1 - a_3/a_2$, with $\lambda_{\min}$ and $\lambda_{\max}$ represent the minimum and maximum eigenvalues of a matrix, respectively. Let $\mathcal{D}_\delta \triangleq \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{x}_{\text{ref}}| < \delta\}$ be a neighborhood of the reference $\mathbf{y}_{\text{ref}}$, with $\delta^2 = (a_3 \Delta_q^2)/(a_1(1 - \rho))$. Here, $\mathbf{x}_{\text{ref}}$ is the reference states that corresponding to the reference outputs $\mathbf{y}_{\text{ref}}$. Suppose that the matrix $\text{diag}(\beta)$ and $\text{diag}(1 - \beta)$ satisfy the relationship $\mathbf{A}_K^T \text{diag}(\beta) \mathbf{A}_K + \mathbf{K}^T \text{diag}(1 - \beta) \mathbf{K} = 0$, here, $\mathbf{A}_K \triangleq \mathbf{A} + \mathbf{B} \mathbf{K}$. In addition, suppose Δ_q is bounded by $((a_1 - a_2\rho)/a_3)(1/|\mathbf{K}|)^2$. Consequently, the states governed by the FCS-MPC will be led to the neighborhood $\mathcal{D}_\delta$, that is

$$\limsup_{k \rightarrow \infty} |\mathbf{x}(k) - \mathbf{x}_{\text{ref}}| \leq \delta \quad (25)$$

for all $\mathbf{x}(0) \in X_{\text{MPC}}$, where $X_{\text{MPC}} \triangleq \mathbb{X}_f \cup \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) < 0\}$.

This stability can be proved by using the Lyapunov function $a_1|\mathbf{x} - \mathbf{x}_{\text{ref}}|^2 \leq V_f(\mathbf{x}) \leq a_2|\mathbf{x} - \mathbf{x}_{\text{ref}}|^2$. Considering the fact that $\mathbf{A}_K^T \text{diag}(\beta) \mathbf{B} + \mathbf{K}^T \text{diag}(1 - \beta) = \mathbf{A}^T \text{diag}(\beta) \mathbf{B} + \mathbf{K}^T (\mathbf{B}^T \text{diag}(\beta) \mathbf{B} + \text{diag}(1 - \beta)) = 0$, it can be confirmed that $V_f(\mathbf{x}(k+1)) \leq \rho V_f(\mathbf{x}(k)) + a_3 \Delta_q^2$. By iterating this inequality for an initial condition $\mathbf{x}(t)$, it follows that:

$$V_f(\mathbf{x}(k)) \leq \rho^k V_f(\mathbf{x}(t)) + \left(\frac{1 - \rho^k}{1 - \rho} \right) a_3 \Delta_q^2.$$

Taking into account the aforementioned Lyapunov function, we can arrive at

$$\tilde{\mathbf{x}}(k) \leq \frac{a_2}{a_1} \rho^k \tilde{\mathbf{x}}(t) + \left(\frac{1 - \rho^k}{1 - \rho} \right) \frac{a_3}{a_1} \Delta_q^2.$$

This way, the practical stability property presented in (25) is proven. For a more rigorous proof of this practical stability of FCS-MPC and the definitions of some aforementioned variables, we refer the reader to [31, Th. 1].

IV. SIMULATION RESULTS

A. Simulation Setup

In order to examine the effectiveness of the proposed control scheme, several sets of simulations are conducted in this section. All simulations are conducted using MATLAB and SimPowerSystems toolbox. The detailed parameters of the DFIG are listed in Table I. The sampling period is set to $T = 10$ ms. The rotor current constraints are chosen to be $i_{rd,\max} = i_{rq,\max} = 300$ A. The block diagram of the simulation system is shown in Fig. 3.

Since the primary objective of the proposed control scheme is to track the active and reactive power references, the control performance is measured by the root-mean-square error (RMSE), which is given by

$$\text{RMSE} = \sqrt{\sum_{k=1}^K \|\mathbf{y}(k) - \mathbf{y}_{\text{ref}}(k)\|^2 / K}. \quad (26)$$

The reactive power reference is given by

$$Q_{\text{ref}} = P_{\text{ref}} \frac{\sqrt{1 - \text{PF}^2}}{\text{PF}} \quad (27)$$

where PF is the reference power factor.

In order to compare the proposed FCS-MPC-BLP scheme with the conventional control scheme, a PI control system is implemented. In this PI control system, a q -axis regulator controls the active power, while a d -axis regulator controls the reactive power. Each regulator has a dual-loop structure, i.e., an inner current loop and an outer power loop. The outputs of the inner loop controllers are limited to $[-489.6 \text{ V}, 489.6 \text{ V}]$, and modulated by the PWM modulator. To make the comparison more fair, the parameters of the PI controllers are optimized using the PID Tuner in the MATLAB Control System Toolbox. Specifically, since the RSC is a nonlinear plant, we first use the PID Tuner to estimate the parameters of its linearized plant model based on time-domain response data. By doing this, for both the d -axis and q -axis regulators, the parameters $K_P = 35.7$ and $K_I = 0.0743$ are selected for the outer loop, and $K_P = 1.6$ and $K_I = 0.0085$ are selected for the inner loop. Finally, the optimized PI controllers are discretized using the Tustin transformation. Furthermore, since the DPC scheme also directly uses the gating signals of the power switches as control inputs [32], the proposed FCS-MPC-BLP is also compared with the DPC. In the DPC control system, we use two three-level hysteresis comparators to generate the active and reactive power states, and use an optimal switching table to generate the optimal control actions, as that in [32].

B. Power Tracking Performance

In this section, we assess the power tracking performance of the proposed FCS-MPC-BLP scheme. The prediction and control horizon is set to $H = 6$ time steps.

First, we keep the rotor mechanical speed at $w_m = 226.6$ rad/s, and test the tracking performance for various active and reactive power steps. In this test, the initial active power reference P_{ref} is set to -60 kW, the initial power factor PF is set to $+0.85$. Hence, the initial reactive power reference

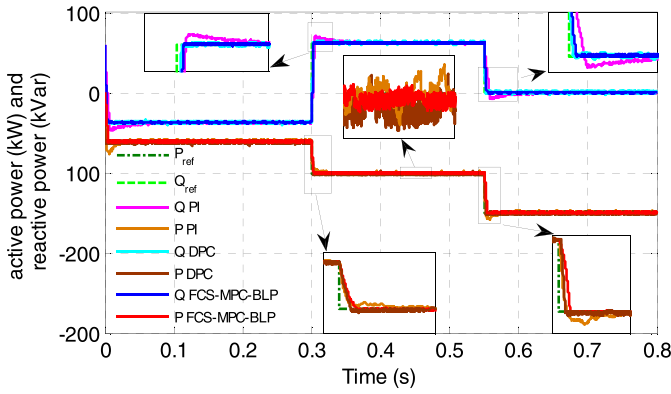


Fig. 4. Response to step changes in the active and reactive power references.

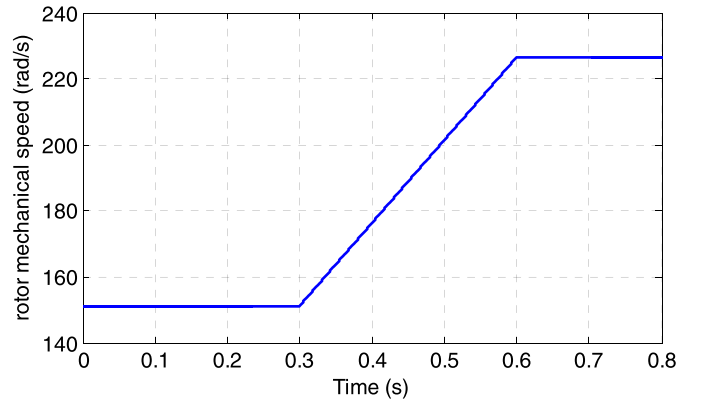


Fig. 6. Variation of the rotor mechanical speed.

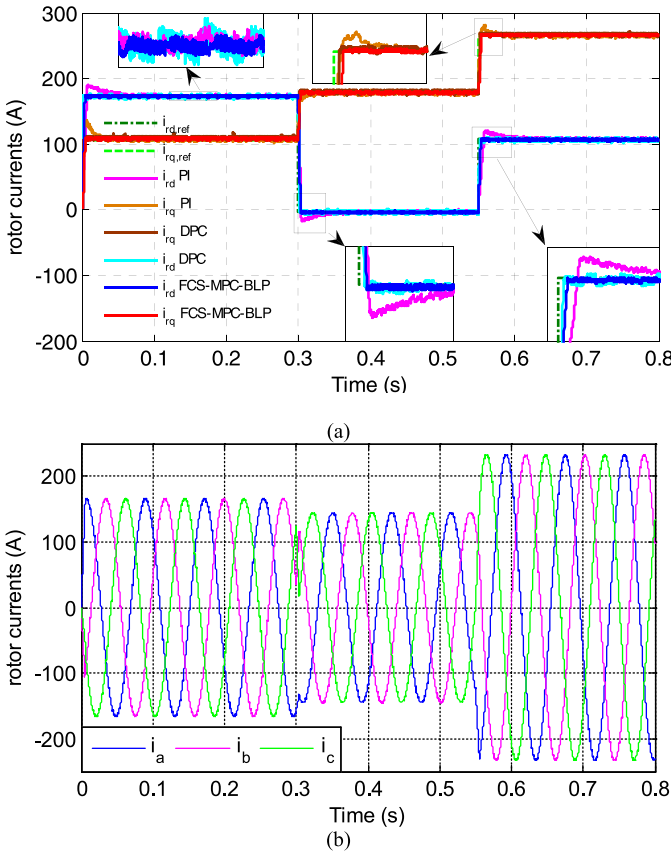


Fig. 5. Rotor current response to the changes in the power references. (a) Synchronous reference frame. (b) Stationary reference frame.

Q_{ref} is -37.18 kVar, due to (27). At 0.3 s, P_{ref} is step changed from -60 kW to -100 kW, and PF is step changed to -0.85 . Correspondingly, Q_{ref} changes to 61.97 kVar. At 0.55 s, P_{ref} is step changed from -100 kW to -149.2 kW, PF changes to 1 . Correspondingly, Q_{ref} changes to 13.29 kVar.

Figs. 4 and 5 present the system responses to these step changes. Fig. 4 shows the step responses of the active and reactive power. From this figure, it is seen that FCS-MPC-BLP clearly yields good tracking performance. Specifically, compared with the PI controller, the FCS-MPC-BLP achieves faster response and smaller overshoot, as shown in the magnified plot. Meanwhile, the steady-state tracking performance is also satisfactory. Compared with the DPC scheme, FCS-MPC-

BLP results in slightly slower dynamic responses during the transient state, due to the fact that DPC directly regulates the DFIG output power. However, the steady-state tracking performance of FCS-MPC-BLP is clearly better than that of DPC. In the magnified plot, one can clearly see that FCS-MPC-BLP provides much smoother power output, both the active and reactive power ripples are much less than those of DPC.

Fig. 5 shows the step response of rotor currents. Fig. 5(a) shows that, with the proposed FCS-MPC-BLP, the rotor currents exhibit good tracking performance with respect to the desired reference trajectories, the dynamic responses during rotor currents steps are within 5 ms. Compared with the PI controller, FCS-MPC-BLP has smaller overshoots. Compared with the DPC, FCS-MPC-BLP presents a similar dynamic response but much less current ripple, as shown in the magnified plot. This can directly result in less ripple of the torque, thus decreasing the unit fatigue and associated maintenance cost.

Second, due to the erratic nature of the atmosphere, the real-time wind speed in wind farm is always changing, rather than constant. Correspondingly, the rotor mechanical speed has to be changed according to the real-time wind speed. Otherwise, the output frequency of DFIG cannot be synchronized with the grid under various wind speed, due to the fact that the pitch control system is quite slow. In light of this, we vary the rotor mechanical speed and test the tracking performance for various active and reactive power steps. To do so, the proposed FCS-MPC-BLP scheme is tested with the rotor speed increasing from 151.1 to 226.6 rad/s during the period 0.3 – 0.6 s, as shown in Fig. 6. Furthermore, at 0.3 s, P_{ref} changes from -60 kW to -100 kW, and PF changes from $+0.85$ to -0.85 . At 0.55 s, P_{ref} changes to -149.2 kW, and PF changes to 1 .

The results of this test are shown in Figs. 7 and 8. From these results, one can see that, under various rotor speeds, the proposed FCS-MPC-BLP still provides satisfactory tracking performance with respect to the desired reference trajectories. Compared with the PI controller, the proposed FCS-MPC-BLP still achieves faster response and smaller overshoot. Compared with the DPC controller, FCS-MPC-BLP still exhibits much better static tracking performance and similar dynamic performance.

Finally, we investigate the effect of the prediction horizon H on the control performance of FCS-MPC-BLP. To achieve

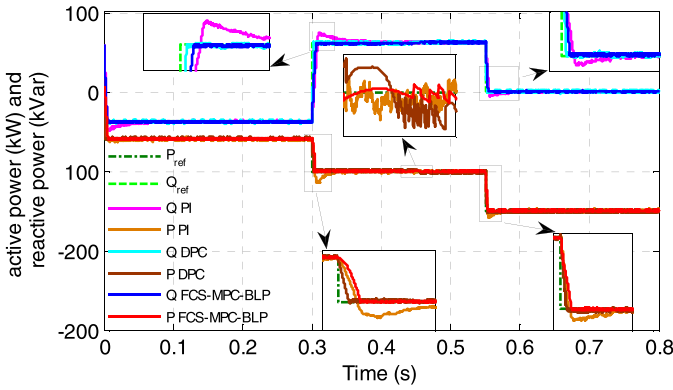


Fig. 7. Response to step changes in the active and reactive power references, under various rotor speeds.

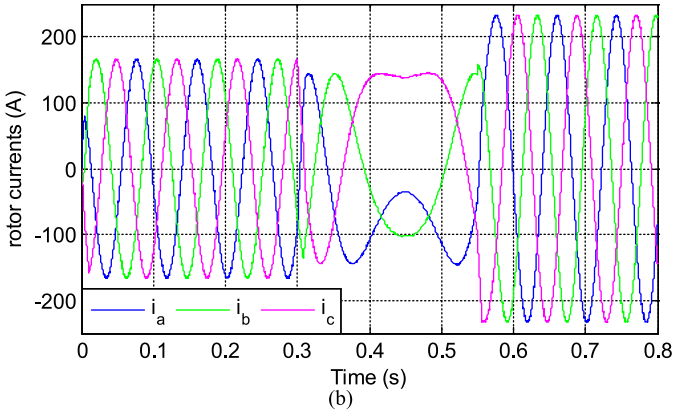
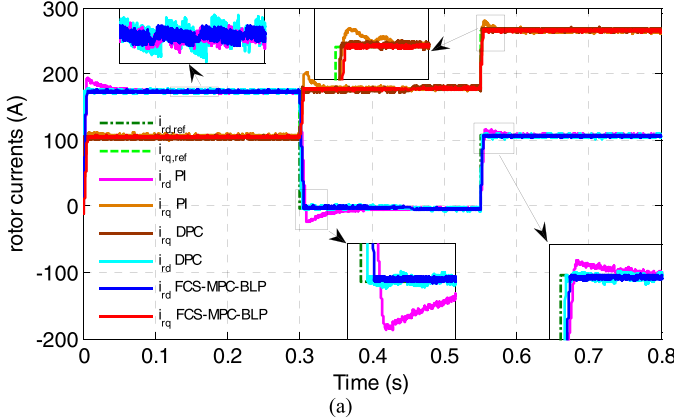


Fig. 8. Rotor current response to the changes in power references and rotor speeds. (a) Synchronous reference frame. (b) Stationary reference frame.

TABLE II
TRACKING PERFORMANCE VERSUS PREDICTION HORIZON

Controller	FCS-MPC-BLP				DPC	PI
H (time step)	1	2	4	6	/	/
RMSE: P (kW)	3.52	3.21	3.13	3.03	3.26	4.41
RMSE: Q (kVar)	5.82	5.71	5.68	5.63	5.67	6.02

this, we repeat the above test with different values of H , and examine its effect on the performance index RMSE. The results of this examination are shown in Table II. From these results, it is clear that longer prediction horizon results in better control performance. In addition, compared with the PI controller, the tracking performance of the FCS-MPC-BLP is significantly better. Compared with the DPC controller,

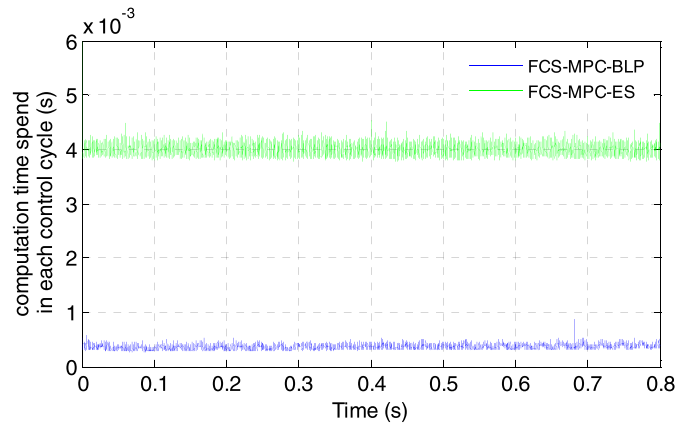


Fig. 9. Evolution of the computation time for each control cycle, $H = 2$ time steps.

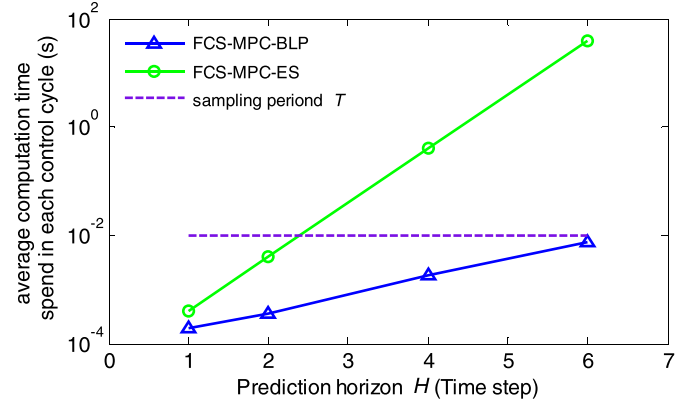


Fig. 10. Average computation time for each control cycle, as a function of prediction horizon.

FCS-MPC-BLP provides better tracking performance when the prediction horizon H is greater than 2 time steps.

C. Computation Time

As analyzed earlier, the key advantage of the proposed FCS-MPC-BLP scheme over the existing FCS-MPC schemes is its ability to reduce computation time. In order to validate this improvement, we compare the computation time of the proposed FCS-MPC-BLP with that of the traditional FCS-MPC, which optimizes the control actions using exhaustive search. To facilitate the subsequent analysis, we denote the traditional scheme as FCS-MPC-ES.

In this simulation, the same test as that of the previous section is conducted for both FCS-MPC-BLP and FCS-MPC-ES, their computation time for each control cycle are then compared. Since the computation time is quite dependent on the prediction horizon H , this simulation is repeated with different values of H . The results of this simulation are shown in Figs. 9–11.

Fig. 9 plots the evolution of computation time for each control cycle. In Fig. 9, the prediction horizon is set to $H = 2$ time steps. It is obvious that FCS-MPC-BLP requires much less computation time than FCS-MPC-ES. In Fig. 10, the average computation time for each control cycle is plotted as a function of H . From Fig. 9, one can see that, with the increase of H , FCS-MPC-ES leads to a striking increase in computation time, the growth rate is much faster than that of

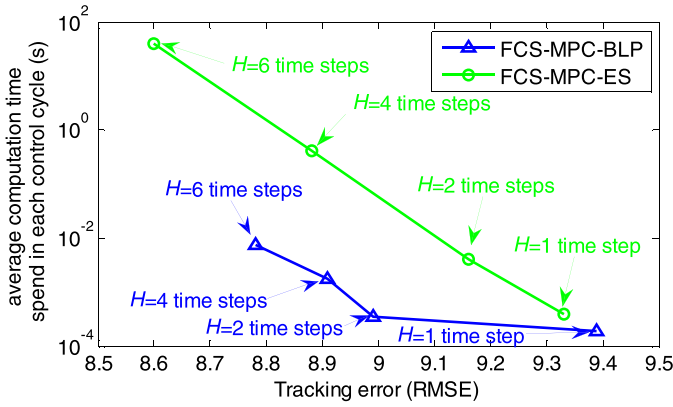


Fig. 11. Control performance versus average computation time for each control cycle.

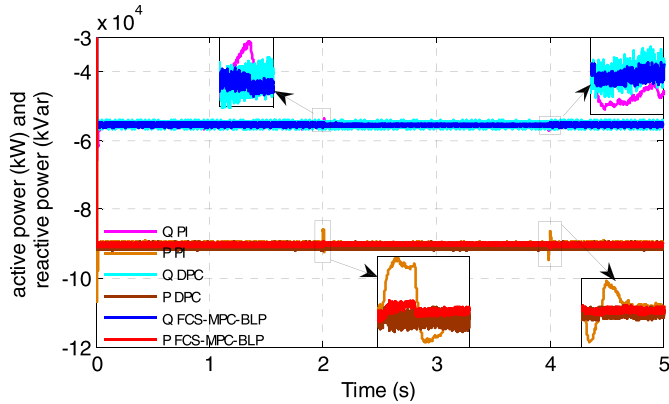


Fig. 12. LVRT performance in terms of active and reactive power outputs.

FCS-MPC-BLP. In addition, for FCS-MPC-ES, when the prediction horizon H is larger than 2 time steps, its computation time for each control cycle will be longer than the sampling period T . As a result, with FCS-MPC-ES, the prediction horizon is restricted to 2 time steps. On the contrary, for FCS-MPC-BLP, even when the prediction horizon H is increased to 6 time steps, the computation time for each control cycle is still shorter than the sampling period.

In Fig. 11, we investigate the tradeoff between computation time and control performance. Here, the control performance is measured by RMSE. From Fig. 11, it is seen that the control performance of FCS-MPC-ES is slightly better than that of FCS-MPC-BLP. However, the computation time of FCS-MPC-ES is much larger. In this sense, compared with FCS-MPC-ES, FCS-MPC-BLP achieves a much better balance between computation time and control performance.

D. Low-Voltage Ride-Through Performance

In this section, we test the low-voltage ride-through (LVRT) performance of the proposed FCS-MPC-BLP scheme. In this test, it is assumed that the fault occurred at 2 s and is cleared at 4 s. During the fault, the stator voltage is suppressed to 66.67% of its nominal value. The rotor speed is set to $\omega_m = 226.6$ rad/s, the stator active and reactive power references are set to $P_{ref} = -90$ kW and $Q_{ref} = -55.8$ kVar, respectively. The results are shown in Figs. 12 and 13. From these results, we see that FCS-MPC-BLP has a much smaller overshoot and better tracking performance than the PI controller. The

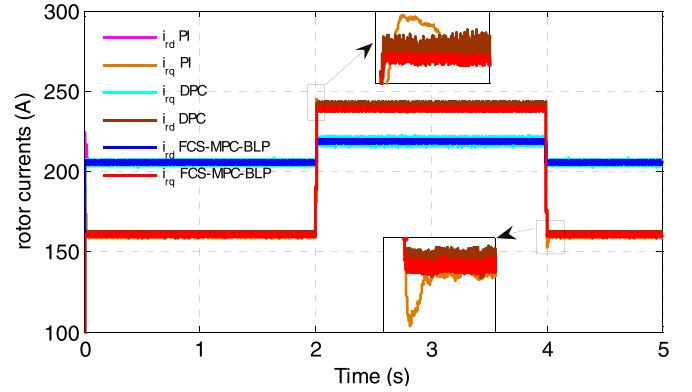


Fig. 13. LVRT performance in terms of rotor currents.

overshoots given by the DPC controller are, in general, similar to those given by the FCS-MPC-BLP. Nevertheless, the power ripple and the current ripple produced by the DPC are much higher than those produced by the FCS-MPC-BLP. This comparison indicates that the proposed FCS-MPC-BLP scheme effectively decreases the over current under the grid disturbance, and hence improving the LVRT performance.

V. CONCLUSION

This paper presents a novel time efficient FCS-MPC-BLP scheme for the power control of DFIG systems. In this scheme, the BQP problem in FCS-MPC is analytically transformed to an equivalent BLP problem, which can be solved efficiently. Simulation results show that, the proposed scheme leads to satisfactory power tracking and LVRT performances, as well as very fast running times. Comparison with the conventional PI controller shows that the proposed scheme achieves better control performance in terms of overshoot, settling time, and disturbance rejection. Comparison with the DPC controller shows that the proposed scheme exhibits better static performance and similar dynamic performance. Comparison with the traditional FCS-MPC (i.e., using exhaustive search) shows that, the proposed scheme significantly reduces the computation time, thus providing a better balance between computation time and control performance. Future works include coordinating the control of RSC and GSC, considering more fault scenarios, and actually implementing the proposed scheme in a field programmable gate array control board. Furthermore, since DPC provides excellent dynamic performance, the combination of DPC and the proposed scheme (i.e., to form a predictive DPC scheme) may further improve the DFIG control performance, the techniques introduced in [14] and [15] may open the possibility to achieve this.

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Peng Kou (S'09–M'13) received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in control science and engineering from Xi'an Jiaotong University, Xi'an, China, in 2005, 2008, and 2013, respectively.

From 2015 to 2016, he visited the Ohio State University, Columbus, OH, USA. He is currently an Associate Professor with the School of Electrical Engineering, Xi'an Jiaotong University. His research interests include renewable energy, predictive control, microgrid, and power system optimization.



Deliang Liang (M'11–SM'15) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 1989, 1992, and 1996, respectively.

Since 1999, he has been with the School of Electrical Engineering, Xi'an Jiaotong University, where he is currently a Professor. From 2001 to 2002, he was a Visiting Scholar with the Science Solution International Laboratory, Tokyo, Japan. His research fields include optimal design, control, and simulation of electrical machines, and electrical machine technology in renewable energy.



Jing Li received the B.S. degree in materials engineering from Xidian University, Xi'an, China, in 2005, and the M.S. degree in electronics science and technology from Xi'an Jiaotong University, Xi'an, in 2008.

She is currently an Engineer with the Academy of Space Information System, Xi'an. Her research interests include the optimization algorithm, electronic ceramics and devices, microwave dielectrics and devices, and dielectric spectroscopy measurement.



Lin Gao received the B.S. and M.S. degrees in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 1986 and 1989, respectively.

She visited Queen's University, Kingston, ON, Canada, from 1993 to 1995. She is currently an Associate Professor with the School of Electrical Engineering, Xi'an Jiaotong University. Her research interests include machine learning, bearingless motor, and magnetic suspension technology.



Qiji Ze received the bachelor's degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2011, where he is pursuing the Ph.D. degree in electrical engineering with the School of Electrical Engineering.

His research interests include motion control, fault tolerant control, doubly salient permanent magnet generator, and power electronics.